# A

# Installation Guide

## A.1 Introduction

The SENCO Support Software is a web-based system designed to help schools manage information about pupils with Special Educational Needs and Disabilities (SEND). It allows staff to store, organise, and access important pupil data in one central location.

The system is intended for use by teachers, SENCO staff, and school administrators. It supports everyday tasks such as recording pupil details, tracking diagnoses, managing records, and logging important events. By bringing all this information together, the system helps staff work more efficiently and make better-informed decisions.

The software is accessed through a web browser, meaning it can be used on different devices such as computers, tablets, or laptops within the school network. It is typically installed on a local server, such as a Raspberry Pi, making it a cost-effective solution for schools. This guide will explain the process using a Raspberry Pi 4.

Due to the sensitive nature of pupil data, the system includes secure login and user access controls to ensure that only authorised users can access information.

## A.2 System Requirements

Before installing the software, ensure the following requirements are met.

### A.2.1 Hardware Requirements

- Raspberry Pi 4 (recommended)
- MicroSD card (at least 16GB)
- Power supply for the Raspberry Pi
- Optional: monitor, keyboard, and mouse (for initial setup only)

The system can also run on a standard computer if preferred.

### A.2.2 Software Requirements

The following software must be installed on the server:

- Raspberry Pi OS (or another Linux-based operating system)
- Apache (web server)
- MariaDB or MySQL (database)
- PHP (version 8 or higher)
- Composer (used to install the application's dependencies)

These components will be installed during the setup process.

### A.2.3 Network Requirements

- The Raspberry Pi must be connected to a local network
- Other devices must be on the same network to access the system
- A web browser (such as Chrome, Edge, or Safari) is required to access the system

### A.2.4 User Requirements

Users should be able to:

- use a web browser
- enter and manage data
- upload files where required

No advanced technical knowledge is needed to use the system once it is installed.

### A.2.5 Security Requirements

- Each user must log in with a username and password
- Access to features is controlled based on user roles
- Only authorised users can view or edit sensitive data

## A.3 Setup and Configuration

This section explains how to set up the system on a Raspberry Pi, and install the application.

### A.3.1 Raspberry Pi Setup

#### Installing the Operating System

1. Download and install Raspberry Pi Imager on your computer.[1]
2. Insert the microSD card into your computer.
3. Open the Imager and select:
    - Device: Raspberry Pi 4
    - OS: Raspberry Pi OS (64-bit)
    - Storage: your microSD card
4. Click **Next** and configure:
    - Host/Device name: SencoServer (or your preference)
    - Localisation (set to your preference)

[1] https://www.raspberrypi.com/software/

- Username and password
- Wi-Fi
- Enable SSH (recommended, using password authentication)
- Raspberry Pi Connect (optional)

5. Click **Write** and wait for the process to complete.
6. Remove the microSD card from your computer and insert it into the Raspberry Pi.
7. Power on the Raspberry Pi.

### Enabling SSH

If SSH was enabled during setup, you can connect to the Raspberry Pi using the device name you created during installation. This will allow you to run commands on the Raspberry Pi, via the terminal on your own computer.

```
ssh username@<device_name>
```

For example:

```
ssh username@SencoServer
```

If this does not work, you can use the device's IP address instead (obtained by running `hostname -I` on the Raspberry Pi):

```
ssh username@<IP_ADDRESS>
```

You may be prompted to continue with the connection. Type `yes` and press Enter. You will then be prompted to enter the password you created during the OS setup.

### Initial Configuration

Update the system:

```
sudo apt update
sudo apt full-upgrade -y
sudo reboot
```

All `sudo` commands may require you to enter the password you created during the OS setup; typing the password might not show any characters on the screen. All commands may

potentially take a few minutes to complete. Ensure that you remain connected to the SSH session throughout the process. Reconnect after reboot.

### A.3.2 Server Setup

#### Installing Apache, PHP, and MariaDB

Install required software:

```
sudo apt update
sudo apt install -y \
    apache2 \
    mariadb-server \
    php \
    libapache2-mod-php \
    php-mysql \
    php-cli \
    php-curl \
    php-mbstring \
    php-xml \
    php-zip \
    php-bcmath \
    php-intl \
    php-gd \
    unzip \
    git \
    composer
```

Enable Apache rewrite module:

```
sudo a2enmod rewrite
sudo systemctl restart apache2
```

#### Cloning the Project

Set permissions:

```
sudo chown -R $USER:$USER /var/www
```

Navigate to the web directory:

```
cd /var/www
```

Clone the project via git, and navigate to the project directory (you can change the directory name 'senco' to your preference, but ensure that you do so in the following commands as well):

```
git clone https://github.com/shafkatkhan/senco-support-software-artefact.git senco
cd senco
```

Install dependencies:

```
composer install --optimize-autoloader
```

Create environment file, and generate application key:

```
cp .env.example .env
php artisan key:generate
```

### A.3.3 Application Setup

#### Creating the Database

Open MariaDB:

```
sudo mysql
```

Run:

```
CREATE DATABASE senco CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
CREATE USER 'senco_user'@'localhost' IDENTIFIED BY 'your_password';
GRANT ALL PRIVILEGES ON senco.* TO 'senco_user'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

**Note:** The values in this query can be customised to your preference, and should be done so before pasting into the terminal:

- Replace `senco` (the database name) with a name of your choice.
- Replace `senco_user` with your preferred database username.

- Replace `your_password` with a secure password of your choice.

Make a note of all three values, as you will need to enter them again when configuring the application in the next step.

### Configuring Apache

Set permissions:

```
sudo chown -R $USER:www-data /var/www/senco
sudo chmod -R 775 /var/www/senco/storage /var/www/senco/bootstrap/cache
```

Create the Apache configuration file:

```
sudo nano /etc/apache2/sites-available/senco.conf
```

Add the following:

```
<VirtualHost *:80>
    ServerName localhost
    DocumentRoot /var/www/senco/public

    <Directory /var/www/senco/public>
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>
```

To save the file in `nano`, press `Ctrl + O`, then press `Enter`. To exit `nano`, press `Ctrl + X`.

Enable the new site configuration:

```
sudo a2dissite 000-default.conf
sudo a2ensite senco.conf
sudo systemctl reload apache2
```

### Completing Setup via Browser

Open a browser on another device connected to the same network, and navigate to your server using the device name you set during installation:

```
http://<device_name>
```

For example:

```
http://SencoServer
```

If this does not work, navigate to the IP address of the Raspberry Pi instead:

```
http://<IP_ADDRESS>
```

The setup wizard will appear. Configure your desired settings, and enter the database credentials you chose earlier:

- Host: `127.0.0.1`
- Port: `3306`
- Database name: the name you chose (e.g. `senco`)
- Username: the username you chose (e.g. `senco_user`)
- Password: the password you set

Once completed, the system is ready to use, and will be accessible from any device on the same network. The default administrator account is created with the username `admin` and the password `password`. It is strongly recommended to change these credentials immediately after the first login.